



Gowin EDP PHY IP User Guide

IPUG1043-2.2E, 07/18/2025

Copyright © 2025 Guangdong Gowin Semiconductor Corporation. All Rights Reserved.

GOWIN is a trademark of Guangdong Gowin Semiconductor Corporation and is registered in China, the U.S. Patent and Trademark Office, and other countries. All other words and logos identified as trademarks or service marks are the property of their respective holders. No part of this document may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, recording or otherwise, without the prior written consent of GOWINSEMI.

Disclaimer

GOWINSEMI assumes no liability and provides no warranty (either expressed or implied) and is not responsible for any damage incurred to your hardware, software, data, or property resulting from usage of the materials or intellectual property except as outlined in the GOWINSEMI Terms and Conditions of Sale. GOWINSEMI may make changes to this document at any time without prior notice. Anyone relying on this documentation should contact GOWINSEMI for the current documentation and errata.

Revision History

Date	Version	Description
03/15/2024	1.0E	Initial version published.
03/29/2024	1.1E	● Descriptions of AFE configuration option added. ● Descriptions of Channel Bonding configuration option added.
06/07/2024	2.0E	Parameters updated.
01/17/2025	2.1E	● Configuration parameters updated. ● Interface screenshots updated.
07/18/2025	2.2E	● IP features updated. ● DRP ports added. ● Configuration parameters updated. ● Interface screenshots updated.

Contents

Contents	i
List of Figures	iii
List of Tables	iv
1 About This Guide	1
1.1 Purpose	1
1.2 Related Documents	1
1.3 Terminology and Abbreviations	1
1.4 Support and Feedback	2
2 Overview	3
2.1 Overview	3
2.2 Features	4
2.3 Resource Utilization	4
3 Functional Description	5
3.1 System Block Diagram	5
3.2 Structure Diagram	5
3.3 Clock	6
3.3.1 Transmit Clock	6
3.3.2 Receive Clock	7
3.4 Data	7
3.4.1 Transmit Data	7
3.4.2 Receive Data	8
3.4.3 Channel Bonding	8
3.5 Port List	8
3.6 Parameter Configuration	11
3.7 Timing Description	14
3.7.1 Gowin EDP PHY TX Interface Timing	14
3.7.2 Gowin EDP PHY RX Interface Timing	14
4 Interface Configuration	15
5 Reference Design	20

5.1 Reference Design Instance 1 20

6 File Delivery 21

6.1 Documentation..... 21

6.2 Design Source Code (Encryption) 21

6.3 Reference Design 21

List of Figures

Figure 3-1 System Block Diagram.....	5
Figure 3-2 Structure Diagram.....	6
Figure 3-3 Gowin EDP PHY IP Port Diagram	9
Figure 3-4 Gowin EDP PHY TX Interface Timing Diagram	14
Figure 3-5 Gowin EDP PHY RX Interface Timing Diagram.....	14
Figure 4-1 Open IP Core Generator	15
Figure 4-2 Open SerDes IP Configuration Interface	16
Figure 4-3 Open EDP PHY Configuration Interface	17
Figure 4-4 EDP PHY Configuration Interface	17
Figure 4-5 General Information Configuration Interface.....	18
Figure 4-6 PHY Configuration	18
Figure 4-7 EDP PHY IP in SerDes IP Configuration Interface	19
Figure 5-1 Reference Design Instance 1 Diagram.....	20

List of Tables

Table 1-1 Terminology and Abbreviations.....	1
Table 2-1 Gowin EDP PHY IP	3
Table 3-1 Gowin EDP PHY IP Port List	10
Table 3-2 Gowin EDP PHY IP Parameter.....	11
Table 6-1 Document List.....	21
Table 6-2 Gowin EDP PHY IP Design Source Code List	21
Table 6-3 Gowin EDP RefDesign Folder Content List.....	21

1 About This Guide

1.1 Purpose

The purpose of Gowin EDP PHY IP User Guide is to help users learn the features and usage of Gowin EDP PHY IP by providing the descriptions of features, functions, ports, timing, GUI and reference design, etc. The software screenshots and the supported products listed in this manual are based on Gowin Software V1.9.11.03 (64-bit). As the software is subject to change without notice, some information may not remain relevant and may need to be adjusted according to the software that is in use.

1.2 Related Documents

The latest user guides are available on the GOWINSEMI website. You can find the related documents at www.gowinsemi.com:

- [DS981, GW5AT series of FPGA Products Data Sheet](#)
- [DS1103, GW5A series of FPGA Products Data Sheet](#)
- [DS1239, GW5AST series of FPGA Products Data Sheet](#)
- [DS1105, GW5AS series of FPGA Products Data Sheet](#)
- [DS1108, GW5AR series of FPGA Products Data Sheet](#)
- [DS1118, GW5ART series of FPGA Products Data Sheet](#)
- [SUG100, Gowin Software User Guide](#)

1.3 Terminology and Abbreviations

Table 1-1 shows the abbreviations and terminology used in this manual.

Table 1-1 Terminology and Abbreviations

Terminology and Abbreviations	Meaning
AFE	Analog Front End
FPGA	Field Programmable Gate Array
IP	Intellectual Property
VS	Vertical Sync
HS	Horizontal Sync

Terminology and Abbreviations	Meaning
DE	Data Enable
VESA	Video Electronics Standards Association
DP	Display Port
EDP	Embedded Display Port
SerDes	Serializer/Deserializer

1.4 Support and Feedback

Gowin Semiconductor provides customers with comprehensive technical support. If you have any questions, comments, or suggestions, please feel free to contact us directly using the information provided below.

Website: www.gowinsemi.com

E-mail: support@gowinsemi.com

2 Overview

2.1 Overview

DisplayPort (DP) is a digital video interface standard developed by the PC and chip manufacturing industry in collaboration with the Video Electronics Standards Association (VESA). This interface is free from certification and licensing fees. It is primarily used for connecting video sources to display devices and supports the transmission of audio, USB, and other forms of data.

The Embedded DisplayPort (eDP) protocol is an extension of the DP protocol specifically designed for embedded systems. eDP is fully compatible with the DP protocol. It is frequently utilized as the interface for liquid crystal displays (LCDs) in tablets and laptop computers.

Gowin EDP PHY IP provides users with a SerDes physical layer access interface that allows users to easily connect the EDP protocol layer to the physical layer.

Table 2-1 Gowin EDP PHY IP

Gowin EDP PHY IP	
Logic Resource	Please refer to 2.3 Resource Utilization
Delivered Doc.	
Design Files	Verilog (encrypted)
Reference Design	Verilog
TestBench	Verilog
Test and Design Flow	
Synthesis Software	GowinSynthesis
Application Software	Gowin Software (V1.9.9 Beta-1 and above)

Note!

For the devices supported, you can click [here](#) to get the information.

2.2 Features

- Supports 1/2/4 lanes
- Supports 8B10B encoding and decoding
- Supports Word Alignment
- Supports 1.62/2.7/5.4/8.1 Gbps/lane
- Supports 16/32/64 bit widths

2.3 Resource Utilization

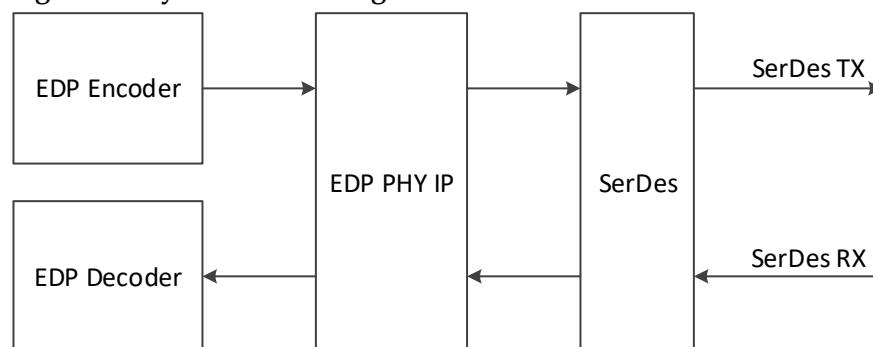
Gowin EDP PHY IP is only used to configure SerDes and does not occupy any Fabric resources.

3 Functional Description

3.1 System Block Diagram

Gowin EDP PHY IP is used to transfer the parallel data encoded by the eDP to the SerDes for conversion to serial data, and to transfer the parallel data to the eDP for decoding after the SerDes receives the serial data and outputs the parallel data. The system block diagram is as shown in Figure 3-1.

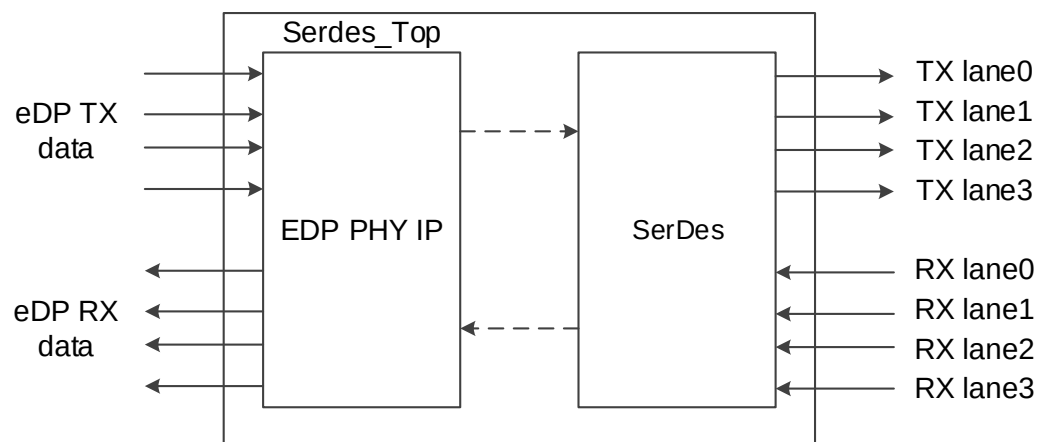
Figure 3-1 System Block Diagram



3.2 Structure Diagram

Gowin EDP PHY IP supports up to four lanes, and each lane includes transmit ports and receive ports. The transmit port receives the K-character signals and data signals from the eDP transmit module and is connected to the SerDes transmit port via port mapping. The receive port connects the output port received by SerDes with the K-character signal and data signal ports of the eDP receive module. Figure 3-2 shows the structure diagram.

Figure 3-2 Structure Diagram



3.3 Clock

3.3.1 Transmit Clock

The transmit clock is generated by the CPLL/QPLL. When configuring the transmit lane, the user needs to configure the speed of the transmit lane, the PLL (CPLL/QPLL) used, as well as the reference clock source and its frequency. Based on the above configuration, the IP configures SerDes PLL to generate a high-speed clock for data transmission. At the same time, SerDes divides the high-speed clock based on user configuration and outputs it as a fabric transmit clock.

- If the user configures the transmit data rate to be 1.62Gbps, and the user clock ratio is selected as 1:20, the Fabric transmit clock is $1.62\text{Gbps}/20=81\text{MHz}$; if the user clock ratio is selected as 1:40, the Fabric transmit clock is calculated as $1.62\text{Gbps}/40=40.5\text{MHz}$; if the user clock ratio is selected as 1:80, the Fabric transmit clock frequency is calculated as $1.62\text{ Gbps}/80=20.25\text{MHz}$.
- If the user configures the transmit data rate to be 2.7Gbps, and the user clock ratio is selected as 1:20, the Fabric transmit clock is calculated as $2.7\text{Gbps}/20=135\text{MHz}$; if the user clock ratio is selected as 1:40, the Fabric transmit clock is calculated as $2.7\text{Gbps}/40=67.5\text{MHz}$. If the user clock ratio is selected as 1:80, the Fabric transmit clock frequency is calculated as $2.7\text{Gbps}/80=33.75\text{MHz}$.
- If the user configures the transmit data rate to be 5.4Gbps, only two clock ratio options are available: 1:40 or 1:80. With a 1:40 ratio, the Fabric transmit clock is calculated as $5.4\text{Gbps}/40=135\text{MHz}$; with a 1:80 ratio, it is calculated as $5.4\text{Gbps}/80=67.5\text{MHz}$.
- If the user configures the transmit data rate to be 8.1Gbps, the only available clock ratio is 1:80, and the Fabric transmit clock is calculated as $8.1\text{Gbps}/80=101.25\text{MHz}$.

When using a 2-lane or 4-lane EDP PHY IP, the fabric_tx_clk_i of each lane shares the pcs_tx_clk_o of one of the lanes.

3.3.2 Receive Clock

When configuring the receive lane, you need to configure the rate of the receive lane. The receive clock is recovered by Clock Data Recovery (CDR), and the CDR outputs the recovered serial data clock for the receive module. At the same time, SerDes divides the serial data clock according to the user configuration and outputs it as a fabric receive clock.

- If the user configures the receive data rate to be 1.62Gbps, and the user clock ratio is selected as 1:20, the Fabric receive clock is calculated as $1.62\text{Gbps}/20=81\text{MHz}$; if the user clock ratio is selected as 1:40, the Fabric receive clock is calculated as $1.62\text{Gbps}/40=40.5\text{MHz}$; if the user clock ratio is selected as 1:80, the Fabric receive clock is calculated as $1.62\text{Gbps}/80=20.25\text{MHz}$.
- If the user configures the receive data rate to be 2.7Gbps, and the user clock ratio is selected as 1:20, the Fabric receive clock is calculated as $2.7\text{Gbps}/20=135\text{MHz}$; if the user clock ratio is selected as 1:40, the Fabric receive clock is calculated as $2.7\text{Gbps}/40=67.5\text{MHz}$; if the user clock ratio is selected as 1:80, the Fabric receive clock is calculated as $2.7\text{Gbps}/80=33.75\text{MHz}$.
- If the user configures the receive data rate to be 5.4Gbps, only two clock ratio options are available: 1:40 or 1:80. With a 1:40 ratio, the Fabric receive clock is calculated as $5.4\text{Gbps}/40=135\text{MHz}$; with a 1:80 ratio, it is calculated as $5.4\text{Gbps}/80=67.5\text{MHz}$.
- If the user configures the receive data rate to be 8.1Gbps, the only available clock ratio is 1:80, and the Fabric receive clock is calculated as $8.1\text{Gbps}/80=101.25\text{MHz}$.

When using a 2-lane or 4-lane EDP PHY IP, `fabric_rx_clk_i` of each lane shares the `pcs_rx_clk_o` of one of the lanes.

3.4 Data

3.4.1 Transmit Data

If the user clock ratio is selected as 1:20, then the transmit parallel data width is 20 bits, including 16-bit data, 2-bit K-character, and 2-bit invalid bits. Only the data `txdata[15:0]` and K-character `txk[1:0]` ports are represented in the IO ports, with the `txk` low bit corresponding to the `txdata` low byte and the `txk` high bit corresponding to the `txdata` high byte.

If the user clock ratio is selected as 1:40, then the transmit parallel data width is 40 bits, including 32-bit data, 4-bit K-character, and 4-bit invalid bits. At the IO ports, only the data `txdata[31:0]` and K-character `txk[3:0]` ports are exposed. The low bits of `txk` correspond to the low bytes of `txdata` and the high bits of `txk` correspond to the high bytes of `txdata`.

If the user clock ratio is selected as 1:80, the transmit parallel data width is 80 bits, including 64-bit data, 8-bit K-character, and 8-bit invalid bits. At the IO ports, only the data port `txdata[63:0]` and the K-character port `txk[7:0]` are exposed. The lower bits of `txk` correspond to the lower bytes of `txdata`, while the higher bits of `txk` correspond to the higher bytes of `txdata`.

SerDes transmit data transmission order: Least Significant Bit (LSB) of

low byte for txdata is transmitted first, and after 8B10B, the data is converted to serial data, and the LSB is transmitted first.

3.4.2 Receive Data

If the user clock ratio is selected as 1:20, then the receive parallel data width is 20 bits, including 16-bit data, 2-bit K-character, and 2-bit invalid bits. Only the data rxdata[15:0] and K-character rxk[1:0] ports are represented in the IO ports, with the rxk low bit corresponding to the rxdata low byte and the rxk high bit corresponding to the rxdata high byte.

If the user clock ratio is selected as 1:40, then the receive parallel data width is 40 bits, including 32-bit data, 4-bit K-character, and 4-bit invalid bits. At the IO ports, only the data rxdata[31:0] and K-character rxk[3:0] ports are exposed. The low bits of rxk correspond to the low bytes of rxdata and the high bits of rxk correspond to the high bytes of rxdata.

If the user clock ratio is selected as 1:80, then the receive parallel data width is 40 bits, including 64-bit data, 8-bit K-character, and 8-bit invalid bits. At the IO ports, only the data rxdata[31:0] and K-character rxk[3:0] ports are exposed. The low bits of rxk correspond to the low bytes of rxdata and the high bits of rxk correspond to the high bytes of rxdata.

SerDes receive data transmission order: rxdata is the parallel data after 8B10B, the LSB is transmitted first, the LSB of each byte is transmitted first.

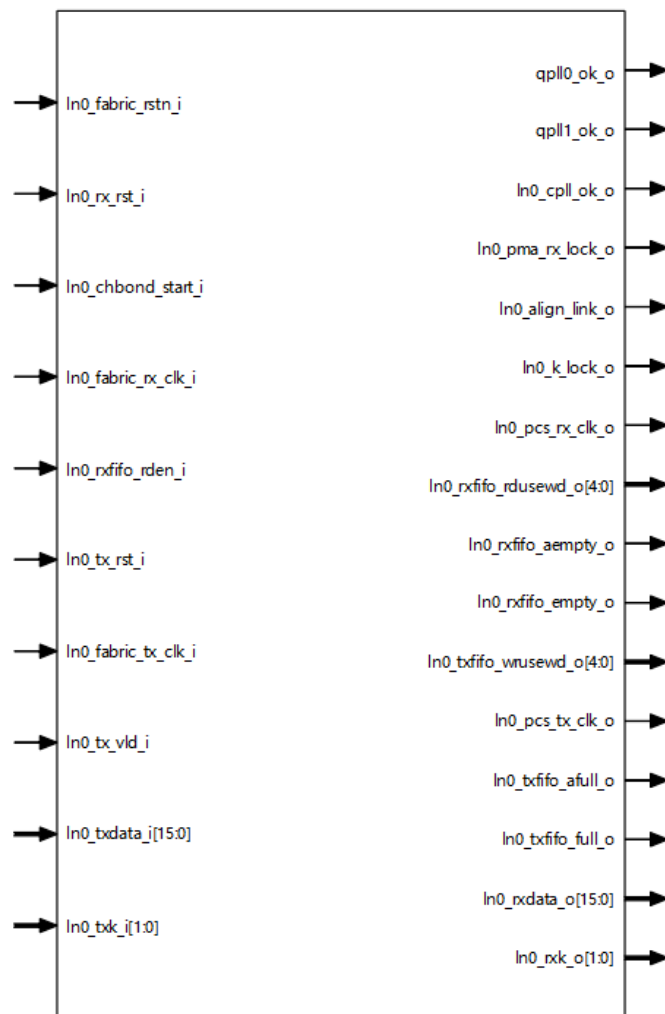
The Word Alignment module can implement receive data alignment. The module aligns the boundary of the receive parallel data according to the K-character K28.5, so that the output parallel data is consistent with the configured K-character.

3.4.3 Channel Bonding

According to the DP protocol, when the number of lanes is greater than 1, there is data skew between lanes. When receiving data, the Channel Bonding option can be enabled to eliminate the data skew between lanes. The Channel Bonding option is set to the One Word, and Pattern0 adopts 0x1C. When all align_link of each Lane is set to 1, the chbond_start control signal of each Lane is enabled. If the Channel Bonding option is not enabled, the RX decoder module must independently eliminate the data skew between lanes

3.5 Port List

The IO ports of Gowin EDP PHY IP are shown in Figure 3-3.

Figure 3-3 Gowin EDP PHY IP Port Diagram

The IO ports vary slightly depending on the parameters.

The details of IO ports of Gowin EDP PHY IP are shown in Table 3-1. Take Lane0 in the table as an example, if other Lane is called, just modify the corresponding port number. Only user ports are listed in the table, and the ports connected to SerDes are connected automatically by Gowin Software.

Table 3-1 Gowin EDP PHY IP Port List

No.	Signal Name	I/O	Description	Note
1	qpll0_ok_o	O	QPLL0 valid indicator 1: valid 0: invalid	The I/O of all the signals takes EDP PHY IP as reference.
2	qpll1_ok_o	O	QPLL1 valid indicator 1: valid 0: invalid	
3	ln0_cppll_ok_o	O	Lane0 CPLL valid indicator 1: valid 0: invalid	
4	ln0_fabric_rstn_i	I	Lane0 fabric reset input, active-low	
5	ln0_rx_rst_i	I	Lane0 RX reset input, active-high	
6	ln0_chbond_start_i	I	Lane0 channel bonding control input, active-high	
7	ln0_pma_rx_lock_o	O	Lane0 RX pma lock indicator 1: Locked 0: Unlocked	
8	ln0_align_link_o	O	Lock indicator for Lane0 RX word align module 1: Locked 0: Unlocked	
9	ln0_k_lock_o	O	Lock indicator for Lane0 RX K-character 1: Locked 0: Unlocked	
10	ln0_pcs_rx_clk_o	O	Lane0 PCS RX clock output	
11	ln0_fabric_rx_clk_i	I	Lane0 RX Buffer read clock, which can be directly connected to edp_ln0_pcs_rx_clk_o	
12	ln0_rxfifo_aempty_o	O	Lane0 RX Buffer almost empty indicator 1: RX Buffer almost empty 0: RX Buffer non-almost empty	
13	ln0_rxfifo_empty_o	O	Lane0 RX Buffer empty indicator 1: RX Buffer empty 0: RX Buffer non-empty	
14	ln0_rxfifo_rdusewd_o	I	Lane0 RX Buffer remaining data indicator	
15	ln0_rxfifo_rden_i	I	Lane0 RX Buffer read enable	
16	ln0_tx_rst_i	I	Lane0 TX reset input, active-high	
17	ln0_pcs_tx_clk_o	O	Lane0 PCS TX clock output	
18	ln0_fabric_tx_clk_i	I	Lane0 TX Buffer write clock, which can be directly connected to edp_ln0_pcs_tx_clk_o	
19	ln0_txfifo_afull_o	O	Lane0 TX Buffer almost full indicator 1: TX Buffer almost full	

No.	Signal Name	I/O	Description	Note
			0: TX Buffer non-almost full	
20	ln0_txfifo_full_o	O	Lane0 TX Buffer full indicator 1: TX Buffer full 0: TX Buffer non-full	
21	ln0_txfifo_wrusewd_o	O	Lane0 TX Buffer remaining data indicator	
22	ln0_tx_vld_i	I	Lane0 TX Buffer read enable	
23	ln0_rxdata_o	O	EDP decoded data	
24	ln0_rxk_o	O	EDP decoded K-character	
25	ln0_txdata_i	I	EDP encoded data	
26	ln0_txk_i	I	EDP encoded K-character	
27	drp_addr_i	I	DRP address	Supported only when the DRP Port is set to Yes; otherwise, they are not supported.
28	drp_wren_i	I	DRP write enable	
29	drp_wrdata_i	I	DRP write data	
30	drp_strob_i	I	DRP write strobe	
31	drp_rden_i	I	DRP read enable	
32	drp_clk_o	O	DRP output clock	
33	drp_ready_o	O	DRP ready indicator	
34	drp_rdvld_o	O	DRP read valid enable	
35	drp_rddata_o	O	DRP read data	
36	drp_resp_o	O	DRP response	

3.6 Parameter Configuration

Table 3-2 Gowin EDP PHY IP Parameter

No.	Name	Range	Default	Description
PHY Configuration				
1	Quad Selection	Q0/Q1	Q0	SerDes Quad selection
2	PHY Lane0	Yes/No	No	SerDes lane0 selection
3	PHY Lane1	Yes/No	No	SerDes lane1 selection
4	PHY Lane2	Yes/No	No	SerDes lane2 selection
5	PHY Lane3	Yes/No	No	SerDes lane3 selection
6	Loopback Mode	OFF/LB_NES/LB_FES/LB_ENC	OFF	Lane loopback mode selection
7	DRP Port	Yes/No	No	Select DRP Port
8	TX Line Rate	1.62/2.7/5.4Gbps	1.62	TX serial rate
9	TX User Clock Ratio	1:20/1:40	1:20	TX user clock ratio
10	TX PN invert Lane0	Yes/No	No	Invert PN on transmit lane 0
11	TX PN invert Lane1	Yes/No	No	Invert PN on transmit lane 1
12	TX PN invert Lane2	Yes/No	No	Invert PN on transmit lane 2
13	TX PN invert Lane3	Yes/No	No	Invert PN on transmit lane 3
14	RX Line Rate	1.62/2.7/5.4Gbps	1.62	RX serial rate
15	RX User Clock Ratio	1:20/1:40	1:20	RX user clock ratio

No.	Name	Range	Default	Description
16	RX PN invert Lane0	Yes/No	No	Invert PN on receive lane 0
17	RX PN invert Lane1	Yes/No	No	Invert PN on receive lane 1
18	RX PN invert Lane2	Yes/No	No	Invert PN on receive lane 2
19	RX PN invert Lane3	Yes/No	No	Invert PN on receive lane 3
20	Reference Clock Source	Q0 REFCLK0 Q0 REFCLK1 Q1 REFCLK0 Q1 REFCLK1 Q0 REFIN0 Q0 REFIN1 Q0 REFIN MCLK	Q0 REFCLK0	Select the reference clock source: - For devices GWAS/T-138 and GWAT-75, the available options are Q0 REFCLK0, Q0 REFCLK1, Q1 REFCLK0, and Q1 REFCLK1. - For GWAT-60 devices, the available options are Q0 REFCLK0, Q0 REFCLK1, Q0 REFIN0, and Q0 REFIN1. - For GWA/T-15 devices, the available options are Q0 REFCLK0, Q0 REFCLK1, and Q0 REFIN/MCLK.
21	Reference Clock Frequency	50-500MHz	135	Reference clock frequency, 135MHz recommended
22	PLL Selection	CPLL QPLL0 QPLL1	CPLL	PLL selection
AFE				
23	Differential Swing	180mV~900mV	900mV	Configure TX differential signal swing Vdiffpp, Vdiffpp=2xVdiff
24	FFE Mode	Auto/Manual	Auto	Configure TX FFE mode - Auto: Automatic mode - Manual: Manual mode
25	Cm	0~19	0	Transmit FFE pre-cursor
26	C0	21~40	40	Transmit FFE main-cursor
27	C1	0~19	0	Transmit FFE post-cursor
28	SD Threshold	25mV~200mV	100mV	Receive differential signal SD threshold
29	Equalization Mode	Auto/Manual	Auto	Configure the receive equalization mode - Auto: Automatic mode - Manual: Manual mode
30	ATT	0~10	7	Adjust the receive intermediate frequency attenuation; the smaller the

No.	Name	Range	Default	Description
				value, the greater the attenuation.
31	BOOST	0~15	9	Adjust the receive high-frequency amplification; the larger the value, the greater the gain
32	BIAS	0~15	7	Configure the amplification parameters of the SerDes for RX signals; the higher the value, the stronger the amplification of the signal.
Channel Bonding				
33	Channel Bonding	None/One Word/Two Words/Four Words	None	RX Channel Bonding enable: ● None: Disable Channel Bonding ● One Word: Enable Channel Bonding for 1 word ● Two Words: Enable Channel Bonding for 2 words ● Four Words: Enable Channel Bonding for 4 words
34	Master Channel Selection	The selected lane	--	Select master channel for receiving Channel Bonding
35	Pattern0	0x00~0xFF	0x7C	The pattern 0 must be a K character
36	Pattern1/2/3	0x00~0xFF	0x7C	The pattern 1/2/3 can be selected as K character
37	K Character	Checked/Unchecked	Unchecked	Configure whether patterns are K characters ● Checked: K character ● Unchecked: Data
38	Max Skew	0~31	8	Configure the maximum skew for RX channels
39	Read Start Depth	0~31	16	After data bonding, configure the read start depth for this module.

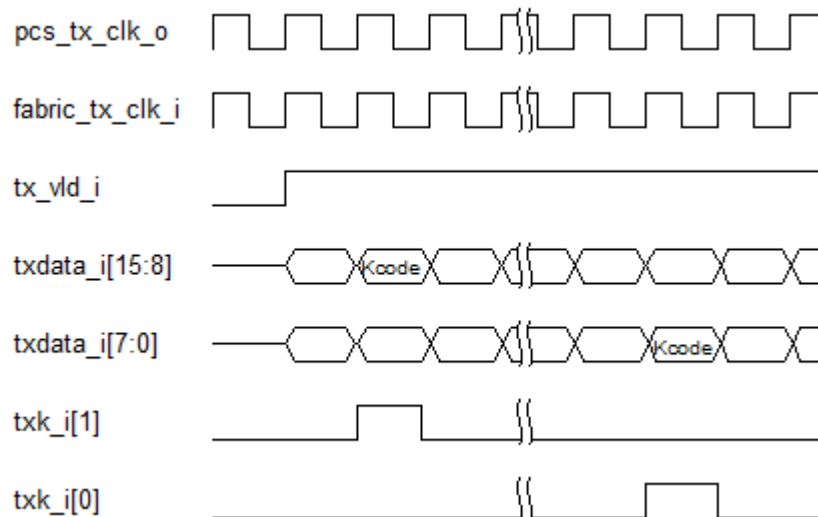
3.7 Timing Description

This section describes the timing of Gowin EDP PHY IP.

3.7.1 Gowin EDP PHY TX Interface Timing

The timing diagram of EDP PHY TX interface is shown in Figure 3-4.

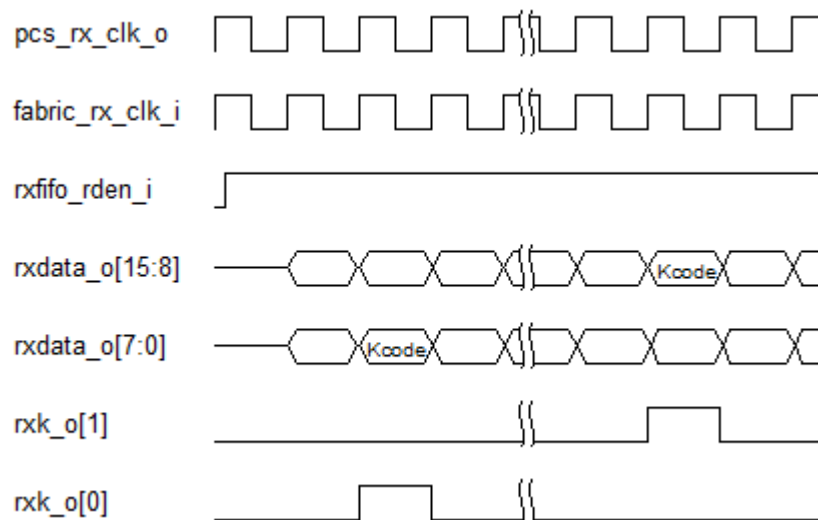
Figure 3-4 Gowin EDP PHY TX Interface Timing Diagram



3.7.2 Gowin EDP PHY RX Interface Timing

The timing diagram of Gowin EDP PHY RX interface is shown in Figure 3-5.

Figure 3-5 Gowin EDP PHY RX Interface Timing Diagram



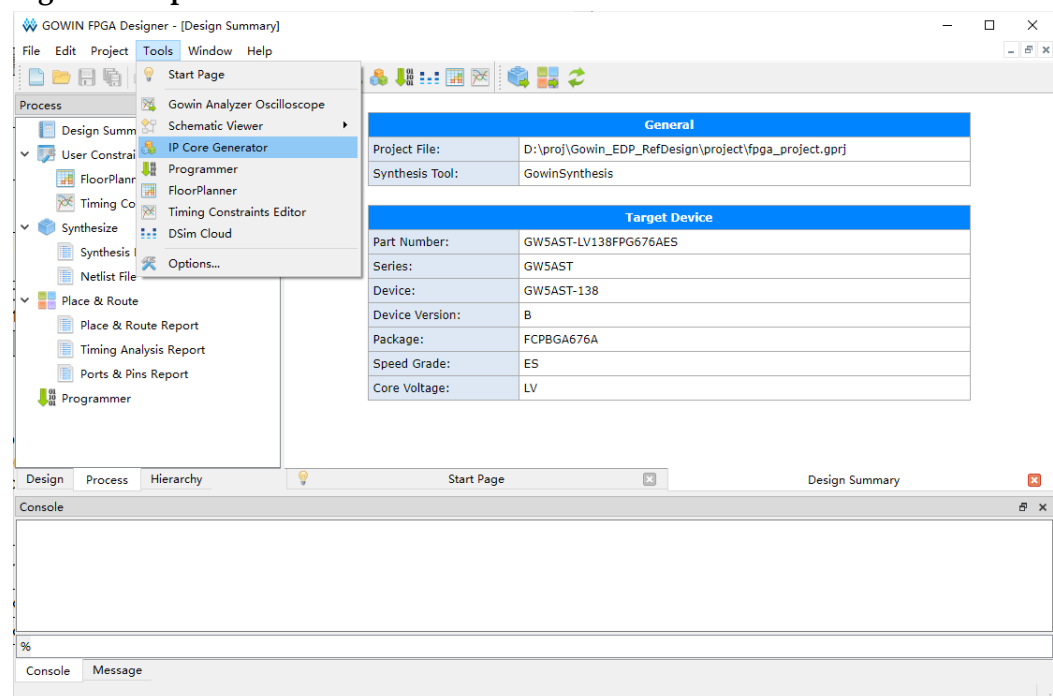
4 Interface Configuration

You can use IP core generator tool in Gowin Software to call and configure Gowin EDP PHY IP.

1. Open IP Core Generator

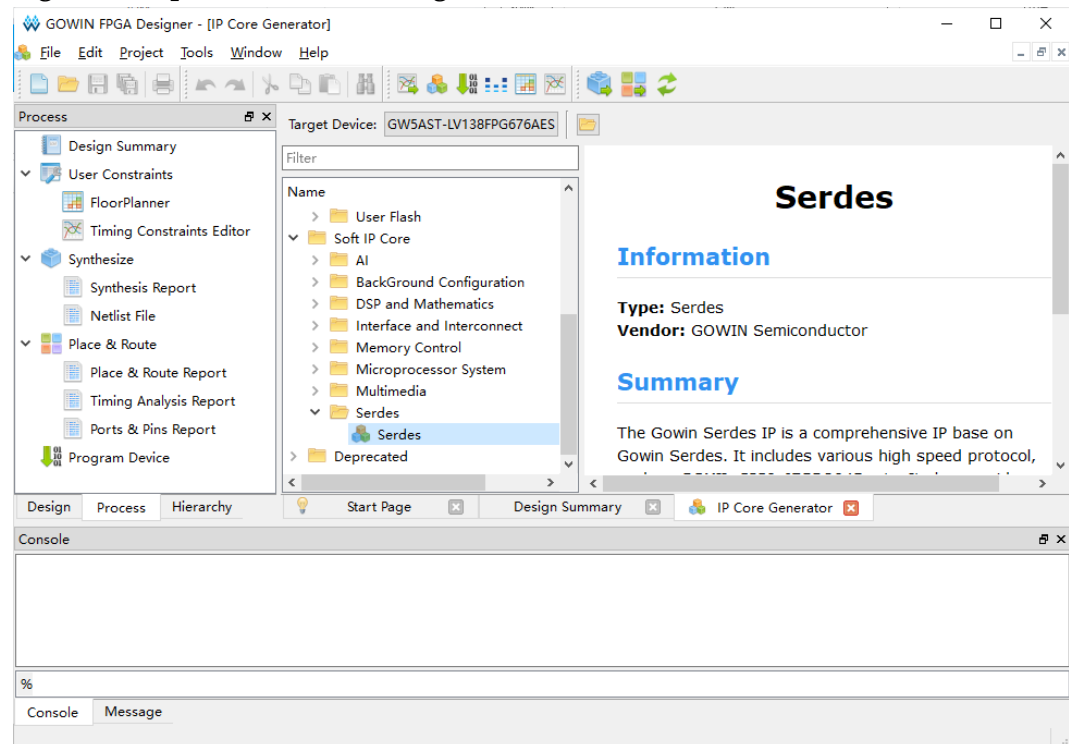
After creating the project, click the "Tools" tab in the upper left, click "IP Core Generator" from the drop-down list to open Gowin IP Core Generator, as shown in Figure 4-1.

Figure 4-1 Open IP Core Generator



2. Open SerDes IP Configuration Interface

Double click "SerDes" to open the configuration interface, as shown in Figure 4-2.

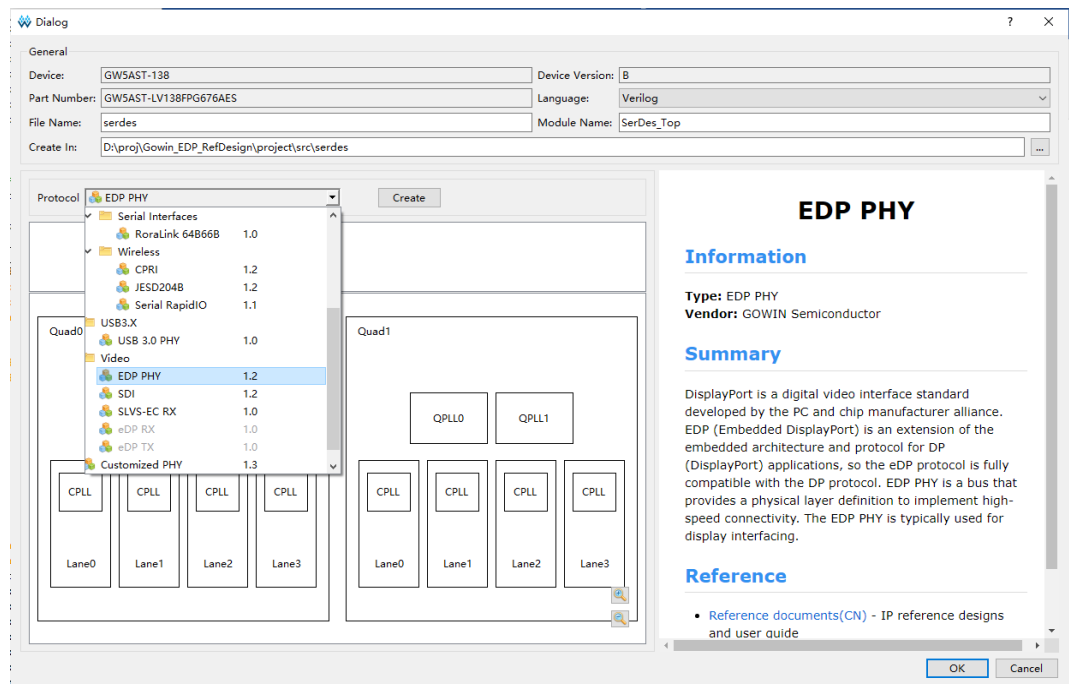
Figure 4-2 Open SerDes IP Configuration Interface

3. Open EDP PHY configuration interface

First configure "General" tab in the SerDes IP interface.

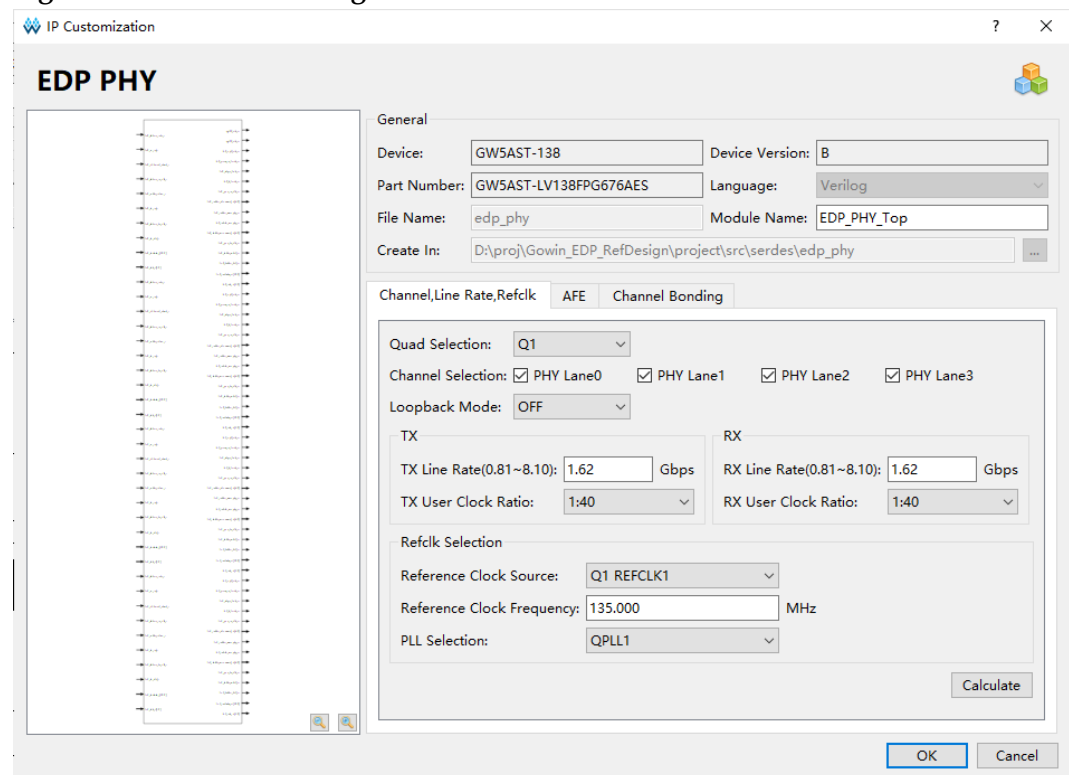
- Device, Device Version, Part Number: Part number settings, determined by the current project, and the user can not set it.
- Language: Support Verilog and VHDL; choose the language as requirements, and the default is Verilog.
- File Name, Module Name, Create In: SerDes file name, module name and the generated file path settings.

Then select the "EDP PHY" option in the "Protocol" option of the SerDes IP configuration interface and click the "Create" button, as shown in Figure 4-3.

Figure 4-3 Open EDP PHY Configuration Interface

4. EDP PHY configuration interface

The configuration interface of Gowin EDP PHY IP is shown in Figure 4-4. The port diagram is on the left, and the parameter configuration options are on the right.

Figure 4-4 EDP PHY Configuration Interface

Configure general information, as shown in Figure 4-5.

See the project general information in the upper section of the configuration interface. The "Module Name" displays the top-level module name of the generated project, and the default is "EDP_PHY_Top"; you can modify the name. The "File Name" displays the folder generated by the IP, which contains the files required by EDP PHY IP, and the default is "edp_phy", and you can modify the path.

Figure 4-5 General Information Configuration Interface

General

Device: GW5AST-138 Device Version: B

Part Number: GW5AST-LV138FPG676AES Language: Verilog

File Name: edp_phy Module Name: EDP_PHY_Top

Create In: D:\proj\Gowin_EDP_RefDesign\project\src\serdes\edp_phy

Configure "PHY Configuration", "AFE", "Channel Bonding" tabs as shown in Figure 4-6.

There is a limitation between the line rate and the reference clock; make sure that the reference clock can generate the above line rate, which can also be detected through clicking "Calculate" button.

Figure 4-6 PHY Configuration

Channel, Line Rate, Refclk AFE Channel Bonding

Quad Selection: Q0

Channel Selection: ☒ PHY Lane0 ☐ PHY Lane1 ☐ PHY Lane2 ☐ PHY Lane3

Loopback Mode: OFF

DRP Port: ☒

TX

TX Line Rate(0.81~8.10): 1.62 Gbps

TX User Clock Ratio: 1:20

TX PN Invert: ☐ Lane0 ☐ Lane1 ☐ Lane2 ☐ Lane3

RX

RX Line Rate(0.81~8.10): 1.62 Gbps

RX User Clock Ratio: 1:20

RX PN Invert: ☐ Lane0 ☐ Lane1 ☐ Lane2 ☐ Lane3

Refclk Selection

Reference Clock Source: Q0 REFCLK0

Reference Clock Frequency: 135.000 MHz

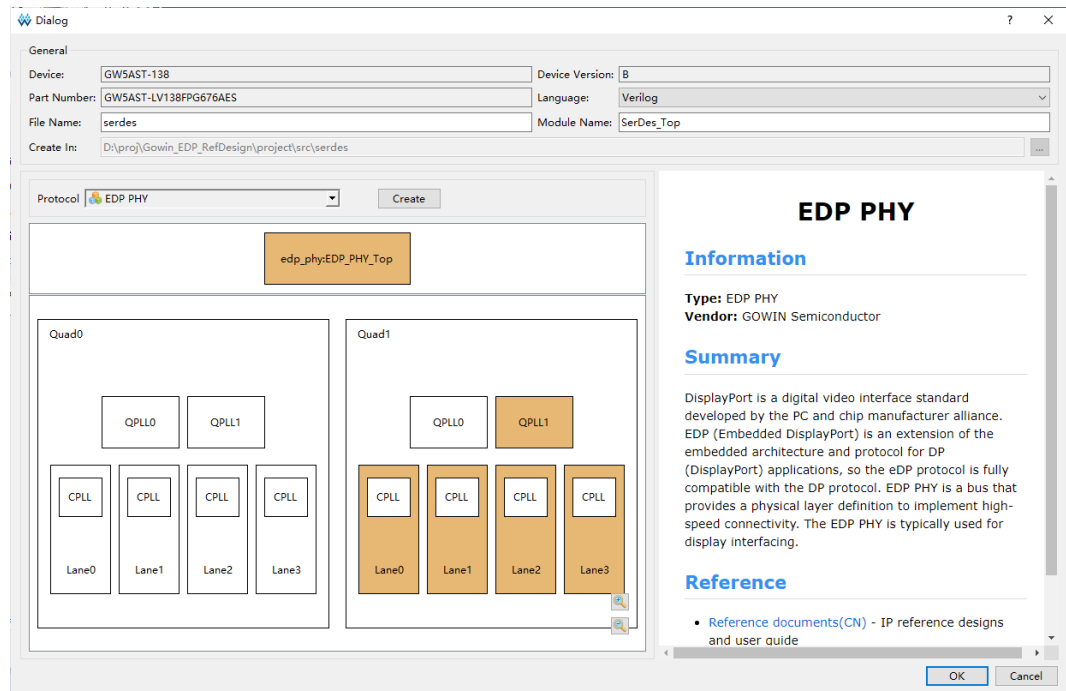
PLL Selection: CPLL

Calculate

5. IP Generation

After completing EDP PHY IP configuration, click "OK" button at the bottom right corner of the interface to generate files of the EDP PHY IP, and return to the SerDes IP configuration interface, and then the SerDes IP configuration interface displays the current generated EDP PHY IP and the corresponding Quad, PLL, and Lane usage, as shown in Figure 4-7.

Figure 4-7 EDP PHY IP in SerDes IP Configuration Interface



Then, click "OK" button at the bottom right corner of the interface to generate SerDes IP files and complete the whole EDP PHY IP generation.

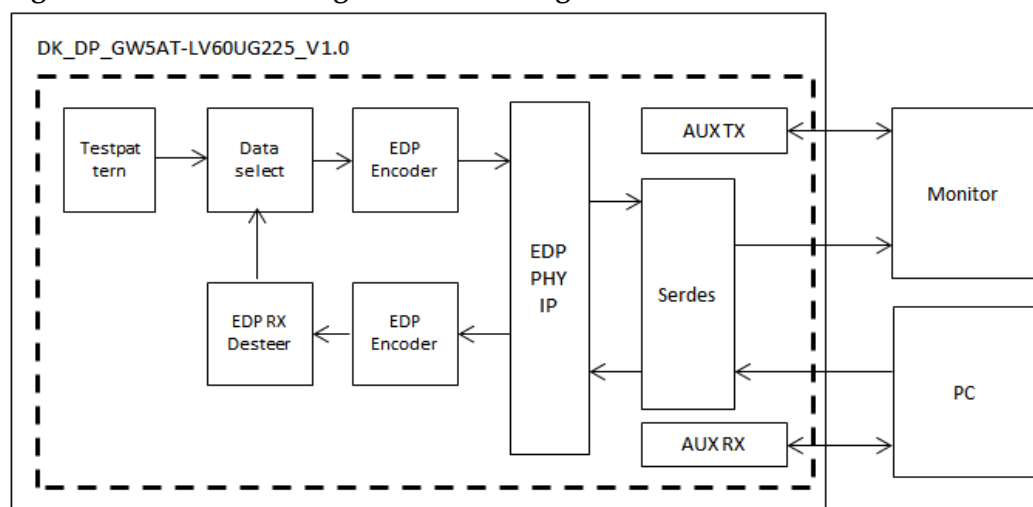
5 Reference Design

This chapter is intended to introduce the usage and structure of the reference design for EDP PHY IP. Please see the [Reference Design](#) for details at Gowinsemi website.

5.1 Reference Design Instance 1

Take DK_CoreBoard_GW5AT-LV138FPG676_V1.0 and DK_DCard_DP-eDP-HDMI-MIPI-LVDS_V1.0 as an example, the diagram is as shown in Figure 5-1. For more information about the development boards, please refer to Gowinsemi website.

Figure 5-1 Reference Design Instance 1 Diagram



In the reference design, there are DP TX and DP RX. The DP TX generates test pattern video signals through Testpattern module; then encode them through EDP Encoder module, input them to EDP PHY IP to convert them into serial signals, and output them to DP display. If no data is received, the DP RX continuously displays the test pattern. For the DP RX, PC outputs DP signals; then convert them to parallel signals through EDP PHY IP, decode them through the EDP Decoder module, convert the format through the EDP RX Desteer module, and then output them to the data selection module before displaying them on the monitor.

6 File Delivery

The delivery files for the Gowin EDP PHY IP includes the documents, the design source code, and the reference design.

6.1 Documentation

The folder mainly contains the user guide in PDF version.

Table 6-1 Document List

Name	Description
IPUG1043, Gowin EDP PHY IP User Guide	Gowin EDP PHY IP User Guide, namely this one.

6.2 Design Source Code (Encryption)

The encrypted code file contains Gowin EDP PHY IP RTL encrypted code which is used for GUI in order to cooperate with Gowin Software to generate the IP core required by users.

Table 6-2 Gowin EDP PHY IP Design Source Code List

Name	Description
edp_phy.v	EDP PHY IP file, encrypted
serdes.v	The top-level file of SerDes-related IP, which provides users with interface information.

6.3 Reference Design

Gowin EDP RefDesign folder contains the netlist file, user reference design, constraints file, top-level file, and project file for Gowin EDP PHY IP.

Table 6-3 Gowin EDP RefDesign Folder Content List

Name	Description
test_top.v	The top module of reference design
serdes	EDP PHY IP folder
testpatternX4.v	Test pattern generation module
key_debounce.v	Key debounce elimination module
key_debounceN.v	Key debounce elimination module
fpga_project.cst	Project physical constraints file

Name	Description
fpga_project.sdc	Project timing constraints file
auxlink	Auxiliary channel AUX communication folder
gowin_pll	PLL IP folder
edp_decoder	EDP receive module folder
edp_encoder	EDP transmit module folder
edp_rx_desteer	EDP RX data sorting module folder

